

Statement of Purpose – Amazon ML Summer School 2026

Raashi Nahata

My journey into AI and Machine Learning began with curiosity about how intelligent systems can perceive, interpret, and act upon real-world information. Over the last two years, I have actively explored this field through projects spanning computer vision, deep learning, signal processing, edge computing, and ML systems engineering. These experiences have helped me move beyond theoretical concepts and understand the challenges of building reliable AI solutions for practical applications.

One of my most significant projects is a Smart Multi-Modal Non-Contact Vital Signs Monitoring System, where I worked on estimating physiological signals such as heart rate using webcam-based remote photoplethysmography (rPPG) along with infrared temperature sensing. Developing this system exposed me to computer vision, signal processing, adaptive sensor fusion, and edge-based deployment. The project has contributed to an institutional patent application under review and an ongoing research effort. Through this experience, I learned the importance of data quality, robustness, and system reliability in real-world ML applications.

I also developed InfraMind, a real-time ML resource monitoring and self-optimization platform that analyzes resource utilization across machine learning workloads and recommends optimization strategies. Working on this project broadened my perspective beyond model development and helped me understand the systems and infrastructure challenges involved in deploying machine learning solutions at scale. The project was selected as a finalist in a national-level 24-hour hackathon.

Additionally, I contributed to SkinSight, a deep learning-based skin disease detection system that strengthened my understanding of model training, dataset preparation, and user-focused AI application development.

Beyond computer vision and applied machine learning, I am currently building an agentic organizational intelligence platform focused on transforming fragmented organizational data into actionable insights. Through this project, I am exploring modern AI concepts such as large language models (LLMs), retrieval-augmented generation (RAG), semantic search, agent-based workflows, and prompt engineering. This project has helped me understand how retrieval and reasoning systems can be combined to solve real-world problems.

While these experiences have provided a strong foundation, I recognize several gaps in my knowledge. I want to deepen my understanding of machine learning theory, probabilistic modeling, optimization techniques, representation learning, and the mathematical principles that drive modern ML systems. I am particularly interested in learning how large-scale production ML systems are designed, evaluated, and continuously improved in industry environments.

Amazon ML Summer School presents a unique opportunity to learn directly from scientists and practitioners who build machine learning systems at scale. The program's structured curriculum, rigorous learning environment, and interaction with experts will help bridge the gap between academic projects and industrial ML practice.

With a strong academic record, hands-on project experience, ongoing research involvement, and a genuine desire to learn, I am excited about the opportunity to contribute to and benefit from the Amazon ML Summer School community. I hope to leverage the knowledge gained through this program to build impactful, scalable, and responsible AI systems in the future.